

Snail Growth Experiment

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2026-09-13

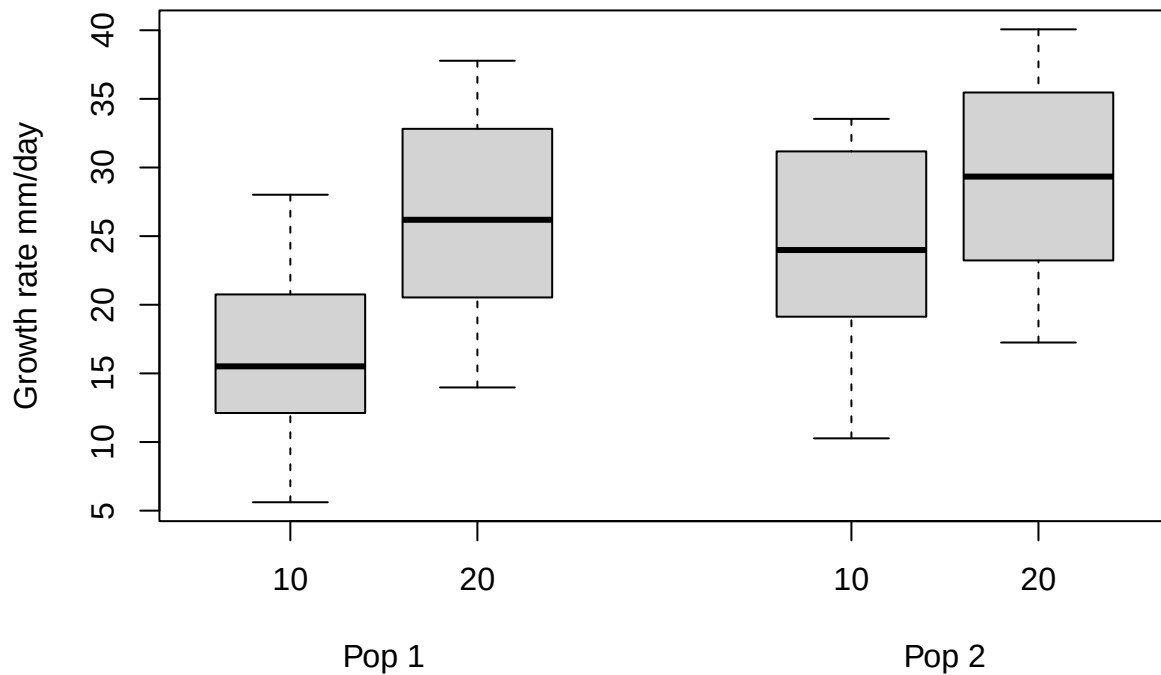
Growth Rate by Temperature and Population

```
df.t <- read.csv('/Users/mcdonaldlab1/Desktop/R/R_week3/tempExperiment_v2.csv')

df.t$temp <- factor(df.t$temp)

boxplot(df.t$growthRate ~ df.t$temp + df.t$pop,
        names = c('10', '20', '10', '20'),
        at = c(1, 2, 4, 5),
        ylab = 'Growth rate mm/day',
        xlab = '')

mtext('Pop 1', side = 1, at = 1.5, line = 3)
mtext('Pop 2', side = 1, at = 4.5, line = 3)
```



ANOVA Results

```
m1.summary <- readRDS('/Users/mcdonaldlab1/Desktop/R/R_week3/m1.summary.rds')
```

```
m1.summary
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## df.t$temp      1  292.8   292.83    3.501  0.0798 .
## df.t$pop        1  125.7   125.75    1.503  0.2379
## df.t$temp:df.t$pop 1   24.3    24.29    0.290  0.5974
## Residuals      16 1338.5    83.65
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

The ANOVA results show that temperature, population, and the interaction between temperature and population were not statistically significant at the 0.05 level. Temperature had a p-value of 0.0798, population had a p-value of 0.2379, and the interaction had a p-value of 0.5974.

Then save this file as: report.rmd